(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:B.Tech.(CSE-AI), Semester:5
(December, 2025)

Subject Code: BAI 305	Subject: Deep Learning-I
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5*4=20)	CO Mapping
	a) "The Perceptron Learning Algorithm can be applied to train a perceptron model for multi-class classification problems". Justify the validity of this statement with suitable reasoning.		CO1
	b) Discuss why L2 regularization is generally considered more applicable than L1 regularization in many deep learning models.		CO2
	c) "TanH activation function is appropriate for the hidden layers of an RNN model." Examine the correctness of this statement with suitable reason.		CO3
	d) Explain the concept of Transfer Learning in context of a DL model along with different techniques for transfer Learning.		CO4
	UNIT I		**CO Mapping**
Q2	a) Explain the MP-neuron (McCulloch–Pitts neuron) model in detail. Further, identify the key advantages and limitations of this computational unit. b) Consider a two-input MP-neuron with excitatory inputs and a threshold value of 1. For the input vectors [-4, –5] and [2, 0.6], determine the corresponding output values produced by the neuron. Additionally, draw the MP-neuron architecture for this scenario.	(10)	CO1
Q3	a) Explain in detail the improvements introduced by the Perceptron unit over the MP-neuron model. b) Discuss why both the Perceptron and the MP-neuron fail to correctly model or classify non-linearly separable Boolean functions.	(10)	CO1
	UNIT II		**CO Mapping**
Q4	Consider a three-input perceptron that uses a sigmoid activation function, with an initial weight vector [2.2, 0.1, 4.0] and a bias of 0.5. The model is trained on two input samples: [7.2, –0.5, 1.2] with target output 1, and [2.0, 1.5, –0.2] with target output 1. Using the cost function as $(y^2\text{-}y^*)^2$, where 'y' is the target output and 'y*' is the estimated output, determine the updated weight values after **three iterations** of the **batch gradient descent** algorithm with a learning rate of **0.1**.	(10)	CO2
Q5	a) Explain the concept of the bias–variance trade-off in the context of Artificial Neural Networks (ANNs). b) Discuss why overfitting tends to be a more prominent concern than underfitting in neural network models, highlighting the key factors that lead to its occurrence.	(10)	CO2

Scanned with OKEN Scanner

	UNIT III		CO Mapping
Q6	a) Discuss five activation functions commonly used in deep learning models, explaining their characteristics and typical applications b) "ReLU activation function is more appropriate for hidden layers of a CNN model." Justify this statement with appropriate reason.	(10)	CO3
Q7	a) Discuss the Backpropogation through time in RNN. b) Explain the LSTM unit along with its advantages and disadvantages.	(10)	CO3
	UNIT IV		**CO Mapping**
Q8	a) "CNN is designed to work upon sequence of inputs while RNN fails to do". Identify the correctness of the above statement with appropriate reason. b) Explain the different weight initialization techniques.	(10)	CO4
Q9	a) Differentiate among ResNet and InceptionNet along with their architecture in detail. b) Explain the 1x1 convolution along with its advantages.	(10)	CO4

Scanned with OKEN Scanner